Task 11 – Direction Sense & Coding-Decoding MCQ Paper

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All 30 practice questions from Task 11, each with four options and exactly one correct answer. Pick one option per question; in `DirectionSense.md` and `CodingDecoding.md`, write the option letter plus your working (coordinates, diagram or rule) – an answer without steps is not accepted.

Part A – Reasoning: Direction Sense (15 questions)

Draw a compass for every question: N up, E right. Right turn = 90° clockwise, left turn = 90° anticlockwise, from the person's own facing direction.

Level 1 – Multi-turn walks

Q1. Riya starts from her home and walks 10 km towards North. She turns right and walks 6 km, turns right again and walks 4 km, then turns left and walks 2 km, and finally turns left and walks 9 km. (i) How far is she from her home, and in which direction? (ii) Which direction is she facing now?

- (a) 15 km North-East; facing East
- (b) 17 km North-West; facing North
- (c) 17 km North-East; facing North
- (d) 13 km North-East; facing West

Q2. A robot starts at point O facing East. It moves 1 m, turns left, moves 2 m, turns left, moves 3 m, and keeps following this pattern (turn left, then move 1 m more than the previous move). After its 8th move, how far is it from O, in which direction, and which way is it facing?

- (a) 4√2 m South-West; facing South
- (b) 4√2 m North-West; facing West
- (c) 8 m South-West; facing South
- (d) 4 m South; facing East

Q3. Starting from point P, Karan walks 12 m West. He turns 90° anticlockwise and walks 5 m, turns 90° anticlockwise again and walks 24 m, then turns 90° clockwise and walks 5 m, turns 90° clockwise again and walks 12 m, and finally turns 90° clockwise and walks 7 m. How far and in which direction is he from P?

- (a) 3 m North
- (b) 7 m South
- (c) 13 m South-West
- (d) 3 m South

Q4. Aman and Bhavna start from the same point. Aman walks 8 km North, turns right and walks 3 km, then turns right again and walks 2 km. Bhavna walks 5 km South, turns left and walks 9 km, then turns left again and walks 3 km. How far apart are they now, and in which direction is Bhavna from Aman?

- (a) 10 km; Bhavna is North-West of Aman
- (b) 10 km; Bhavna is South-East of Aman
- (c) 14 km; Bhavna is South-East of Aman
- (d) 8 km; Bhavna is South of Aman

Level 2 – Degrees, shadows and clocks

Q5. Neeraj is facing North-East. He turns 135° clockwise, then 270° anticlockwise, and then 45° clockwise. Which direction is he facing now?

- (a) South-West
- (b) North-East
- (c) North-West
- (d) South-East

Q6. One evening, just before sunset, Sameer and Tanya were standing face to face in a park. Tanya's shadow fell exactly to Sameer's left. Which direction was Sameer facing?

- (a) South
- (b) North
- (c) East
- (d) West

Q7. A clock is kept on a table so that at 3:00 PM its minute hand points towards North-East. In which direction will the hour hand point at 7:30 PM?

- (a) South-West
- (b) South
- (c) North-West
- (d) West

Q8. Pooja walks 20 m towards East, turns left and walks 20 m. She then turns 135° clockwise and walks 20√2 m. How far and in which direction is she from her starting point?

- (a) 20√2 m South-East
- (b) 40 m East
- (c) 20 m East
- (d) 40 m North-East

Level 3 – Relative positions

Q9. Point Q is 6 km East of point P. R is 8 km North of Q. S is 6 km West of R. T is 4 km South of S. U is 3 km East of T. (i) In which direction is U from P, and how far? (ii) What is the distance between U and Q?

- (a) (i) 5 km North-East; (ii) 5 km
- (b) (i) 5 km North-East; (ii) 4 km
- (c) (i) 7 km North-East; (ii) 5 km
- (d) (i) 5 km North-West; (ii) 3 km

Q10. Six friends A, B, C, D, E and F stand in a ground. A is 10 m West of B. C is 10 m South of B. D is 10 m East of C. E stands exactly midway between A and D. F is 5 m East of E. (i) In which direction is F from C, and how far? (ii) Who stands exactly midway between B and C?

- (a) (i) 5 m North; (ii) E
- (b) (i) 5√2 m North-West; (ii) F
- (c) (i) 5√2 m North-East; (ii) E
- (d) (i) 10 m North-East; (ii) D

Q11. Village Q is North of village P. Village R is East of Q. Village S is West of P. Village T is South of S. No distances are given. In which direction is R from T? (Also explain in your answer file why the answer does not depend on the distances.)

- (a) North
- (b) North-East
- (c) East
- (d) Cannot be determined without distances

Level 4 – Coded directions

Code	Meaning

`P $ Q`	P is 5 m North of Q
`P # Q`	P is 5 m East of Q
`P @ Q`	P is 5 m South of Q
`P % Q`	P is 5 m West of Q

Q12. If `A $ B # C @ D`, in which direction is A from D, and how far?

- (a) 5 m North
- (b) 5√2 m North-East
- (c) 5√2 m South-East
- (d) 5 m East

Q13. Which expression shows that X is to the North-East of Z?

- (a) `X $ Y % Z`
- (b) `X # Y $ Z`
- (c) `X @ Y # Z`
- (d) `X % Y @ Z`

Level 5 – Rotated compass

Q14. If North-East becomes West, and all other directions change in the same manner, what will South become?

- (a) North-East
- (b) North-West
- (c) East
- (d) South-West

Q15. In a strange town, directions are renamed so that North is called South-East, and all other directions are renamed in the same manner. A man walks 10 km towards the direction the town calls "West", then turns left and walks 10 km. In real directions, how far and in which direction is he from his starting point?

- (a) 20 km East
- (b) 10√2 km South-East
- (c) 10√2 km East
- (d) 10√2 km North

Part B – Aptitude: Coding-Decoding (15 questions)

Find the rule on the first 2–3 letters, then check it holds for every letter before you choose.

Level 1 – Letter coding

Q1. In a certain code, BRAIN is written as CTDMS. How is SMART written in that code?

- (a) TNDVY
- (b) TODVY
- (c) TODUY
- (d) UODVY

Q2. If GARDEN is written as OFESBH, how is FLOWER written in that code?

- (a) SFXPMH
- (b) GMPXFS
- (c) SEXPMG
- (d) SFXPMG

Q3. In a certain code, WATER is written as DZGVI. How is SMILE written in that code?

- (a) HNROV
- (b) HMROV
- (c) HNRPV
- (d) GNROV

Q4. In a certain code, MOTHER is written as NMUFFP. How is FATHER written in that code?

- (a) GZUFFP
- (b) EYUFFP
- (c) GYUFFP
- (d) GYUGFP

Q5. If COMPUTER is written as PMOCRETU, how is KEYBOARD written in that code?

- (a) DRAOBYEK
- (b) BYEKDRAO
- (c) BYEKDROA
- (d) KEYBDRAO

Level 2 – Number coding

Q6. If PAINT is coded as 74128 and EXCEL is coded as 93596, how is ACCEPT coded?

- (a) 455978
- (b) 455987
- (c) 545978
- (d) 455798

Q7. In a certain code, ACE is written as 1925 and BAD is written as 4116. How is FIG written?

- (a) 698
- (b) 364981
- (c) 368194
- (d) 368149

Q8. If CAT = 72, DOG = 78 and FISH = 168, then LION = ?

- (a) 150
- (b) 208
- (c) 200
- (d) 50

Level 3 – Sentence, substitution and decoding

Q9. In a certain code language, 'sky is blue' is written as 'ta na pa', 'blue and green' is written as 'ka ta ma', and 'green is fresh' is written as 'na ma ra'. What is the code for 'sky'?

- (a) pa
- (b) ta
- (c) na
- (d) ma

Q10. If 'pen' is called 'pencil', 'pencil' is called 'eraser', 'eraser' is called 'sharpener', and 'sharpener' is called 'scale', what will a student use to rub out a pencil mark?

- (a) eraser
- (b) sharpener
- (c) scale
- (d) pencil

Q11. In a code, every letter is replaced by the letter 3 places after it in the alphabet (X → A, Y → B, Z → C). Decode this message: KDVK PDS

- (a) NGYN SGV
- (b) GASH MAP
- (c) HASH MAT
- (d) HASH MAP

Level 4 – Conditional coding (Q12–Q15)

Letters are coded using this table:

Letter	A	B	D	E	G	K	M	P	R	T
Code	`5`	`@`	`3`	`#`	`9`	`1`	`%`	`7`	`$`	`2`

Conditions:

1. If the first letter is a vowel and the last letter is a consonant, the codes of the first and the last letters are interchanged.
2. If both the first and the last letters are consonants, both are coded as `*`.
3. If the first letter is a consonant and the last letter is a vowel, the last letter is coded as the code of the first letter.

If no condition applies, code every letter as per the table.

Q12. EGPKMT

- (a) `#971%2`
- (b) `*971%*`
- (c) `2971%#`
- (d) `2971%2`

Q13. MBRADK

- (a) `*@$53*`
- (b) `%@$531`
- (c) `1@$53%`
- (d) `*@$53%`

Q14. TGAKPE

- (a) `29517#`
- (b) `295172`
- (c) `#95172`
- (d) `*9517*`

Q15. AKDTPE

- (a) `#13275`
- (b) `513275`
- (c) `*1327*`
- (d) `51327#`